School of Engineering

E211 – Operations Planning II

Lesson 16 Worksheet

Shortest Path Problem (Dijkstra's algorithm)

1. Referring to distribution routes below (Figure 1), draw a network diagram of the possible delivery routes from Johor Bahru to each of the other cities. What do the **nodes** represent in this network? What do the **arcs** represent in the network?

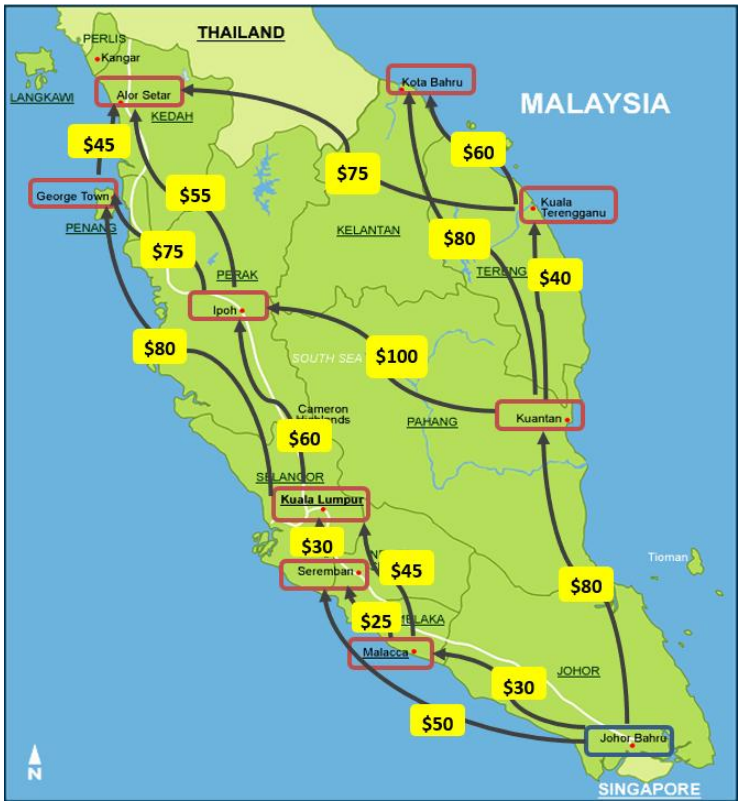

Figure 1: WantTheToys Unit Cost Distribution Routes in Malaysia

Nodes represent the cities in this network. Arcs represent the delivery route cost of delivering toys from Johor Bahru (JB) to 9 other cities.

2. List down all possible routes that can transport the toys from Johor Bahru to Alor Setar. How many possible routes are there? () represent the total cost to deliver

   Johor Bahru – Malacca – Kuala Lumpur – George Town – Alor Setar (200)

   Johor Bahru – Malacca – Kuala Lumpur – Ipoh – George Town – Alor Setar (255)

   Johor Bahru – Malacca – Kuala Lumpur – Ipoh – Alor Setar (190)

   Johor Bahru – Malacca – Seremban – Kuala Lumpur – George Town – Alor Setar (210)

Johor Bahru – Malacca – Seremban – Kuala Lumpur – Ipoh – George Town – Alor Setar (265)
Johor Bahru – Malacca – Seremban – Kuala Lumpur – Ipoh – Alor Setar (200)

Johor Bahru – Seremban – Kuala Lumpur – George Town – Alor Setar (205)
Johor Bahru - Seremban – Kuala Lumpur – Ipoh – George Town – Alor Setar (260)
Johor Bahru – Seremban – Kuala Lumpur – Ipoh – Alor Setar (195)

Johor Bahru – Kuantan – Ipoh – George Town – Alor Setar (300)
Johor Bahru – Kuantan – Ipoh – Alor Setar (235)
Johor Bahru – Kuantan – Kuala Terengganu – Kota Bahru (180)
Johor Bahru – Kuantan – Kuala Terengganu – Alor Setar (195)
Johor Bahru – Kuantan – Kota Bahru (160)

There are possible 14 routes.

3. Out of all the possible routes from Johor Bahru to Alor Setar, which route do you think has the **least** cost for delivering one truckload of toys? What type of problem is this and what is one approach you can use to solve this problem?
   Least cost: Johor Bahru – Kuantan – Kota Bahru (160).
   This is **shortest path problem** and one approach to solve this problem is using **Dijkstra's algorithm.**

4. Applying Dijkstra's algorithm, work out the delivery routes that would achieve the least cost for delivering the toys **from Johor Bahru to each of the other 9 cities**. You may use the following template to work out your solution.

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L16 Templates.xlsx

Iteration	Solution Set	Arcs	Total Cost
1	JB	**JB - M** JB - S JB - K	**30** 50 80

Iteration	Solution Set	Arcs	Total Cost
2	JB M (30)	**JB - S** JB - K M - S M - KL	**50** 80 30+25=55 30+45=75

Iteration	Solution Set	Arcs	Total Cost
3	JB	JB - K	80
	M (30)	**M - KL**	**30+45=75**
	S (50)	S - KL	50+30=80

Iteration	Solution Set	Arcs	Total Cost
4	JB	**JB - K**	**80**
	M (30)	KL - GT	75+80=155
	S (50)	KL - IPOH	75+60=135
	KL (75)		

Iteration	Solution Set	Arcs	Total Cost
5	JB	K - IPOH	80+100=180
	M (30)	**K - KT**	**80+40=120**
	S (50)	KL - GT	75+80=155
	KL (75)	KL - IPOH	75+60=135
	K (80)	K - KB	80+80=160

Iteration	Solution Set	Arcs	Total Cost
6	JB	KL - GT	75+80=155
	M (30)	**KL - IPOH**	**75+60=135**
	S (50)	K - IPOH	80+100=180
	KL (75)	KT - KB	120+60=180
	K (80)	KT - AS	120+75=195
	KT (120)	K - KB	80+80=160

Iteration	Solution Set	Arcs	Total Cost
7	JB	K - KB	80+80=160
	M (30)	IPOH - GT	135+75=210
	S (50)	IPOH - AS	135+55=190
	KL (75)	**KL - GT**	**75+80=155**
	K (80)	KT - KB	120+60=180
	KT (120)	KT - AS	120+75=195
	IPOH (135)		

Iteration	Solution Set	Arcs	Total Cost
8	JB	GT - AS	155+45=200
	M (30)	IPOH - AS	135+55=190
	S (50)	KT - AS	120+75=195
	KL (75)	KT - KB	120+60=180
	K (80)	**K - KB**	**80+80=160**

	KT (120) IPOH (135) GT (155)		

Iteration	Solution Set	Arcs	Total Cost
9	JB M (30) S (50) KL (75) K (80) KT (120) IPOH (135) GT (155) KB (160)	GT - AS **IPOH - AS** KT - AS	155+45=200 **135+55=190** 120+75=195

Iteration	Solution Set	Arcs	Total Cost
10	JB M (30) S (50) KL (75) K (80) KT (120) IPOH (135) GT (155) KB (160) AS (190)		

SHORTEST PATH SOLUTION		
From JB to	Shortest Path	Total Cost
M	JB -> M	30
S	JB -> S	50
KL	JB -> M -> KL	75
K	JB -> K	80
KT	JB -> K -> KT	120
IPOH	JB -> M -> KL -> IPOH	135
GT	JB -> M -> KL -> GT	155
KB	JB -> K -> KB	160
AS	JB -> M -> KL -> IPOH -> AS	190

5. What are some assumptions made when deriving the solution to the shortest path problem?
   Total cost of shipment is proportional to the amount shipped
   The shipment follows a single, continuous path from origin to destination

Unit cost of shipment is fixed.

Assumptions: (From 6P Solution)

- Unit cost of shipment does not change with shipment quantity.
- Routes in the network are fixed.
- There is no limitation of route capacity.

Maximum Flow Problem (Ford-Fulkerson's algorithm)

6. Refer to Figure 2, draw the network diagram of the toy delivery routes clearly indicating the Origin and Destination nodes. Indicate the flow capacity along each arc.

Figure 2: WantTheToys Distribution Routes Capacity in Malaysia

Pick any delivery route from Johor Bahru to Alor Setar. What is the maximum quantity of toys (number of truckloads) that can be delivered along that route?

Johor Bahru – Malacca – Kuala Lumpur – George Town – Alor Setar

Max qty: 50

7. Using Ford-Fulkerson's algorithm, determine the maximum quantity of toys (in number of truckloads) that WantTheToys can deliver from Johor Bahru to Alor Setar. You may make use of the template in question 4.

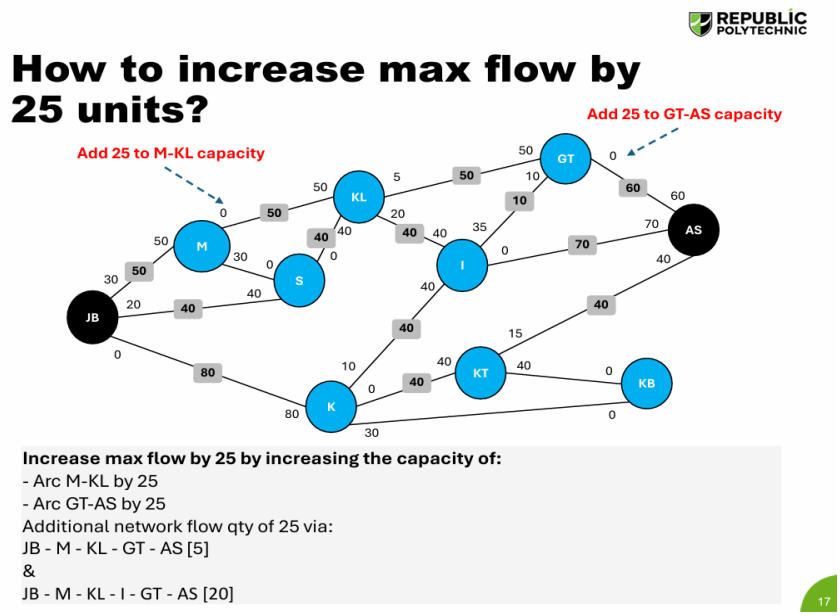
Route	Flow Quantity
JB - M - KL - GT - AS	50
JB - S - KL - I - AS	40
JB - K - KT - KB	40
JB - K - KB	30
JB - K - I - GT - AS	10
Total	**170**

8. What are some assumptions when deriving the solution to the maximum flow problem?
   Cost of delivery is not a factor in determining the routes
   Products are only delivered from Johor to Alor Setar
   Delivery routes and capacity are fixed

9. There is a request for WantTheToys to deliver another 25 truckloads of toys from Johor Bahru to Alor Setar in addition to the max quantity determined above. Determine where the extra delivery capacity should be added and the quantity to add.
   Arc M-KL by 25
   Arc GT-AS by 25

   JB - M - KL - GT - AS [5], JB - M - KL - I - GT - AS [20]

   Answer Key:



10. Would the algorithms you apply to solve the shortest path and maximum flow problems today – Dijkstra's algorithm and Ford-Fulkerson's algorithm – always give optimal solutions?

    Both algorithms give optimal solutions.